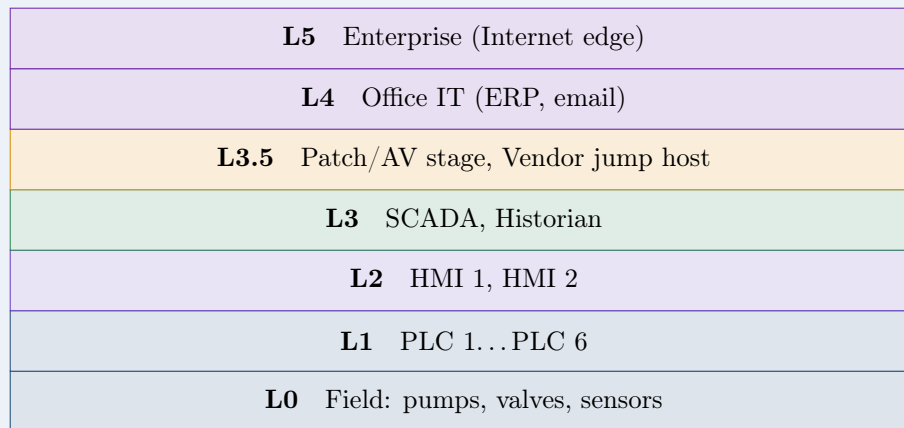


Industrial Security (Bachelor)

Solution Sheet – Section 03 Industrial Network Architectures

Exercise 3.1 – Purdue Diagram (sample)



Vendor remote support terminates in the iDMZ on the jump host, never directly in L1/2.

Exercise 3.2 – Zones & Conduits (sample)

Zones:

- Z1 Field (SL-T 2), Z2 Control (SL-T 3), Z3 Supervisory (SL-T 3), Z4 iDMZ (SL-T 2), Z5 IT (SL-T 1).

Conduits (excerpt):

- C1 Z3→Z4 (Historian replication, OPC UA, push-only, TCP/4840).
- C2 Z5→Z4 (Historian read for BI, REST/HTTPS).
- C3 Z4→Z3 (vendor jump-host RDP, MFA, time-bound).

A data diode is appropriate for C1: it is push-only and integrity of the BPCS must not be at risk from any IT-side compromise. A stateful firewall still depends on software correctness; a diode enforces direction in hardware.

Exercise 3.3 – Lab Notes

Expected outcome: `nmap -Pn 192.168.10.0/24` from the Enterprise host returns no open ports. Replication test (e.g. `nc -lvp 5450` on the DMZ) succeeds only from the Control subnet.

Exercise 3.4 – Topology Trade-Offs

	Star	Ring (HSR/MRP)
Failure modes	Switch is single point of failure	Survives single link failure
Recovery time	Manual switch replacement	Sub-second (HSR is 0 ms)
Cabling cost	Lowest	Higher; needs ring-capable switches